

Timur Mudarisov

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Applied Scientist / Research Engineer | LLMs, Transformer Analysis, Applied ML

Profile

PhD candidate and machine learning researcher working on LLMs, attention, and applied ML, with two accepted NeurIPS papers (2025, 2026), prior experience in financial-model validation and time-series forecasting, and research spanning theory, experimentation, and applied NLP. Strong foundation in Python/C/C++ and probability/statistics, with hands-on experience building reproducible PyTorch/Hugging Face evaluation workflows for decoder-only LLMs up to 9B parameters.

Experience

University of Luxembourg, SnT – SEDAN

Mar 2024 – Present

Doctoral Researcher, Luxembourg

- Design and evaluate machine-learning methods for LLMs, attention mechanisms, NLP, and quantitative-finance applications.
- Translate research questions into empirical experiments and publish results at venues including NeurIPS 2025, NeurIPS 2026, ICNLSP, and ACM ICAIF.
- Collaborate on applied projects across explainable ML, NLP, and quantitative finance.
- Build reproducible PyTorch/Hugging Face experimentation and evaluation pipelines for transformer analysis, perturbation studies, low-rank compression, structured pruning, and knowledge distillation across 1B–9B decoder-only LLMs.
- Implement matrix-free model diagnostics using Jacobian-vector/vector-Jacobian products, finite differences, and Fisher/Hessian approximations to quantify output sensitivity, effective rank, and prediction-relevant hidden directions.
- Maintain public experimental artifacts and reviewer-follow-up analyses for LLaMA-2-7B and Mistral-7B on OpenWebText, including layerwise effective-rank, finite-difference, and Fisher/Hessian comparisons.

Raiffeisen Bank

Sep 2022 – Jan 2024

Moscow, Russia

- Validated financial models including rates and mortgage models.
- Applied machine-learning methods to time-series forecasting.
- Assessed model assumptions, predictive stability, and validation evidence for rates, mortgage, and time-series models, with emphasis on robust out-of-sample evaluation.

WorldQuant

Sep 2021 – Dec 2021

Quantitative Researcher Intern, Moscow, Russia

- Applied financial models for return prediction.
- Developed and evaluated return-prediction hypotheses using quantitative research workflows, statistical validation, and out-of-sample model comparison.

New Economic School

Mar 2022 – May 2022

Teaching Assistant, Moscow, Russia

- Taught modern machine learning methods.

Selected Research and Impact

- **Predictive Geometry of Hidden Trajectories in Transformers – NeurIPS 2026 (accepted).** [code/results] Introduced a pullback-Fisher framework for loss-to-go geometry in decoder-only Transformers, separating output-sensitive from prediction-null hidden directions and deriving tokenwise predictive saliency. Evaluated geometry-guided rank allocation and structured pruning on ten 1B–9B LLMs across WikiText, OpenWebText, and FineWeb; geometry-regularized distillation improved reverse-KL recovery in 9/14 and skew-KL recovery in 10/14 matched settings.
- **Limitations of Normalization in Attention Mechanism – NeurIPS 2025.** Derived non-asymptotic distance and geometric-separability bounds for softmax-type attention together with a Jacobian bound for normalization stability. GPT-2 experiments confirmed long-context attention dilution, 70–85% separability saturation, and the predicted $1/T$ gradient-sensitivity trade-off.
- **Cross-Sector Market Regime Forecasting with LLM-Augmented News Analysis – ACM ICAIF 2024.** FinBERT ensemble + time-series baseline comparison; reported 73% higher accuracy and 110% higher F1 than a straightforward time-series baseline.

- **Scalable Text Vectorization with Hyperdimensional Computing Through Selective Word Encoding – ICNLSP 2025.** Introduced compression HDC (CHDC), combining hyperdimensional text encoding with TF-IDF/LDA word selection; derived estimators and bounds for compression rate, Jensen–Shannon divergence, and ROUGE, and validated them on IMDB, arXiv, and AG News. TF-IDF compression reduced encoded text size to 10% or less in some settings.

Selected Engineering / Open-Source Projects

- **Utility-based Market Making** – Python project exploring utility-based approaches to market making.
- **Limit Order Book Dynamics** – Notebook-based implementation inspired by limit-order-book models and optimal trading strategies.
- **Heston Model Implementation** – Implementation of the Heston model for option-price prediction.
- **Predictive Geometry – Experimental Supplement** – Public reviewer-follow-up results for LLaMA-2-7B and Mistral-7B on OpenWebText, with layerwise Fisher/Hessian, effective-rank, finite-difference, and future-leakage analyses.

Education

University of Luxembourg – SnT <i>PhD Candidate, Doctoral School in Science and Engineering, Luxembourg</i> Thesis: <i>A Geometric and Operator-Theoretic Framework for Transformer Mechanism.</i> Develops a unified geometric and operator-theoretic view of attention, multi-head structure, feed-forward transformations, residual-stream dynamics, and prediction-relevant hidden-state geometry.	Mar 2024 – Expected Mar 2027
New Economic School <i>Master in Economics and Data Science</i>	Sep 2020 – Jul 2022 GPA: 4.2/5.0
Yandex School of Data Analysis <i>Economics and Data Science Program</i>	Sep 2020 – Jul 2022
Moscow State University <i>Specialist, Mechanics and Mathematics – Probability Theory Department</i>	Sep 2016 – Jul 2022 GPA: 4.5/5.0

Technical Skills

Programming	Python, C, C++
ML	LLMs, Transformers, attention, NLP, model compression/distillation, time-series forecasting, quantitative ML, statistics
Libraries	PyTorch, Hugging Face Transformers
Research Engineering	Reproducible experimentation, model evaluation, ablation studies, statistical validation, JVP/VJP and finite-difference diagnostics, GPU-based LLM analysis, Git/GitHub
Applied Scientist	Problem formulation, experimental design, rigorous benchmarking, technical communication, cross-functional research collaboration
Languages	English (B2), French (A2), German (A1)